FastSpeech 2 TTS for English and Telugu: Technical Report

Name: Sreya Sudheeran
College: COEP Technological University, Pune
Github: https://github.com/SreyaSudheeran/fastspeech2-english-telugu

1. Implementation approach

I built two separate FastSpeech 2 models, one for English and one for Telugu, by adapting ming024's open-source PyTorch implementation (github.com/ming024/FastSpeech2) of the FastSpeech 2 paper (Ren et al., 2020). The model, loss and optimiser stay as in the reference. The work went into the text frontends, data preparation, alignment, logging and inference tooling for each language.

Components reused unchanged from the reference

Component	Setting
Encoder and decoder	4 encoder and 6 decoder feed-forward Transformer (FFT) blocks, hidden size 256, 2 attention heads, convolutional filter size 1024 with kernel sizes (9, 1).
Variance adaptor	Separate duration, pitch and energy predictors (filter size 256, kernel 3, dropout 0.5), a length regulator, and 256 linearly spaced bins for the pitch and energy embeddings.
Loss	L1 on the mel-spectrogram before and after the PostNet; MSE on log-duration, pitch and energy.
Optimiser	Adam (0.9, 0.98) with the Transformer warm-up schedule (4,000 warm-up steps), gradient clipping at 1.0, batch size 16.
Targets	Phoneme durations from the Montreal Forced Aligner (MFA); pitch from the WORLD vocoder (DIO + StoneMask); energy as the L2 norm of the STFT magnitude. Pitch and energy are averaged per phoneme and z-normalised with statistics from the training split only.
Vocoder	The universal HiFi-GAN generator shipped with the reference repository, used as is. Audio is 22.05 kHz, 80 mel bins, hop 256, window 1024, mel range 0 to 8 kHz.

The resulting model has about 35.4 million parameters.

Modifications

Area	Change	Why
Symbol table	Chosen per language with one switch. English uses ARPAbet phones with stress marks; Telugu uses a new 46-phone inventory. Pinyin symbols removed.	Each model needs its own vocabulary.
Telugu front-end (new)	Unicode normalisation; expansion of numbers, times and ordinals into Telugu words (for example 2026 becomes the words for "two thousand twenty six"); rule-based grapheme-to-phoneme conversion that handles the inherent vowel, virama, vowel signs and nasal assimilation of the anusvara.	Telugu spelling is close to phonemic, and no pretrained Telugu G2P was assumed. The same code writes the aligner dictionary and converts test sentences, so training and inference phones agree.
English front-end	Context-aware G2P using part-of-speech tags and a small noun/verb heteronym table (for example CONduct vs conDUCT); clock-time and AM/PM expansion; one pronunciation dictionary built for the aligner.	The reference looks words up without context, so it cannot separate the two readings of "conduct" in the test set.

Dataset tooling (new)	Transcript filtering, deduplication, round-robin subsampling across speakers, train/validation split, resampling, dictionary export.	Needed to get about 5 h of train and 1 h of validation audio from each corpus.
Feature extraction	Split lists replace the random shuffle. Pitch and energy statistics come from training data only. Silences that MFA 3 writes as empty labels are kept and treated as pauses. Utterances over the length limit are skipped.	Without the empty intervals every pause disappears from the durations, and timing drifts after the first pause.
Sequence limit	Maximum length raised from 1,000 to 1,500 mel frames (about 17 s).	Clips of up to 15 s are used.
Training loop	CSV loss logs, best-validation checkpoint, resume from the latest checkpoint, TensorBoard made optional.	Loss plots for the report; robustness to Colab disconnects and a TensorBoard/TensorFlow clash.
Compatibility	librosa 0.10 and current PyTorch calls; STFT no longer hard-coded to CUDA.	The reference pins 2020-era library versions that do not install on current Colab.
Inference	One script per language that synthesises the test sentences and plots predicted duration, pitch and energy.	Required deliverables.

English-specific

Svarah has no speaker id, so I derived a pseudo-speaker id by hashing each recording's speaker metadata (gender, age group, native state and district, education, occupation). That gives 115 pseudo-speakers. The English model is therefore multi-speaker: a 256-dimensional speaker embedding is added to the encoder output. The reported audio uses the pseudo-speaker with the most training data.

Telugu-specific

One speaker was selected, the one with the most usable audio, so the Telugu model is single-speaker. No pretrained Telugu aligner was assumed, so an MFA acoustic model was trained on the selected Telugu data itself.

2. Datasets, preprocessing and training setup

Each model uses 5.0 h (English) or 4.7 h (Telugu) of training audio and about 1 h of validation audio, with no overlap between the two. The Telugu training set falls slightly under 5 h because 5.5 h of audio was selected and alignment then trimmed the silence at the start and end of each clip.

Selection and subsampling

Both corpora were read from Hugging Face. Transcripts were normalised per language. A clip was dropped if its transcript contained annotation tags or characters outside the language's alphabet, if it was shorter than 1 s or longer than 15 s, or if its transcript duplicated another clip's (one random clip was kept per sentence, so train and validation share no sentence). Clips were then drawn round-robin across speakers (seed 1234) up to the target hours plus a safety margin (20% for English, 10% for Telugu) to absorb alignment failures, split by utterance into train and validation, and finally trimmed to the target hours after alignment.

Step	English (Svarah)	Telugu (IndicTTS Telugu)
Source	ai4bharat/Svarah, 6,656 clips, 9.6 h, 117 speakers, test split only	SPRINGLab/IndicTTS_Telugu, 8,576 clips
Dropped: bad transcript	1,253	138
Dropped: duration outside 1 to 15 s	353	3,185
Dropped: duplicate transcript	186	13
After filtering	5,050 clips, 7.80 h	5,253 clips, 11.80 h
Speaker policy	115 pseudo-speakers from speaker metadata (Svarah has no speaker id); multi-speaker model	Two voices with 3.74 h and 8.06 h of usable audio; the larger one kept; single-speaker model
Selected before alignment	train 3,797 clips (5.88 h), validation 844 clips (1.32 h)	train 2,567 clips (5.50 h), validation 518 clips (1.10 h)
Aligned	4,641 of 4,641	3,084 of 3,085
Final training set	3,457 clips, 5.00 h, 115 speakers	2,566 clips, 4.69 h, 1 speaker
Final validation set	698 clips, 1.00 h, 114 speakers	518 clips, 0.94 h, 1 speaker
Pitch statistics (training set)	mean 179.7 Hz, std 57.3 Hz	mean 187.4 Hz, std 42.6 Hz

Train and validation separation

A check on the final feature files found no shared utterance id and no shared transcript between train and validation in either language, and every validation speaker also appears in training. Clips with identical audio were excluded during data preparation (the preparation step asserts that no audio is shared).

Preprocessing

Audio was resampled to 22.05 kHz and peak-normalised. Phoneme durations came from forced alignment, and leading and trailing silence was trimmed. Pitch was extracted with WORLD, interpolated over unvoiced frames and averaged per phoneme. Energy is the STFT magnitude norm averaged per phoneme. Both were z-normalised with training-set statistics.

Alignment

English used the pretrained MFA 3.4.3 model `english_us_arpa` with a dictionary of all CMUdict pronunciation variants for the 7,248 words in the data.

For Telugu, an MFA acoustic model was trained on the selected clips (3,075 utterances, 6.58 h, 46 phones; mean log-likelihood -46.2) using the rule-based Telugu dictionary, and then used to align them. MFA's TextGrid export crashed on this setup (an assertion on zero-length intervals), so alignments were exported as JSON and converted to TextGrids with a small script. The script dropped 309,894 exact duplicate intervals (MFA had stored every alignment twice after reusing its working database) and repaired 143 overlapping intervals in 28 files. The workaround is documented in the repository README.

Training configuration

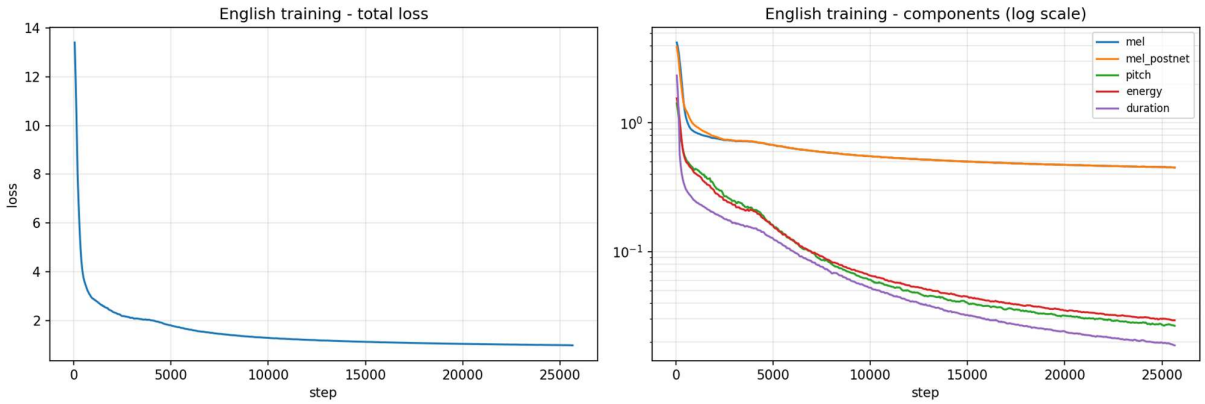
Setting	**English**	**Telugu**
Hardware	Google Colab free T4 GPU	Google Colab free T4 GPU
Batch size, optimiser	16, Adam (0.9, 0.98), warm-up 4,000 steps, gradient clip 1.0	Same
Configured schedule	60,000 steps	10,000 steps (drops at 7,000, 8,500, 9,500)
Steps actually trained	About 25,000, then stopped (GPU quota)	4,000, then stopped (GPU quota)
Validation / checkpoints	Validation every 1,000 steps; checkpoint every 2,000 steps plus best-validation	Validation every 1,000 steps; checkpoint every 2,000 steps plus best-validation
Checkpoint used for results	Best validation loss, step 12,000 (total loss 2.539)	Step 4,000 (lowest mel and post-net mel loss; best total validation loss was at step 2,000, 2.232)
Feature storage during training	Google Drive	Copied to the Colab local disk (Drive reads were too slow for many small files)

3. Training and validation loss analysis

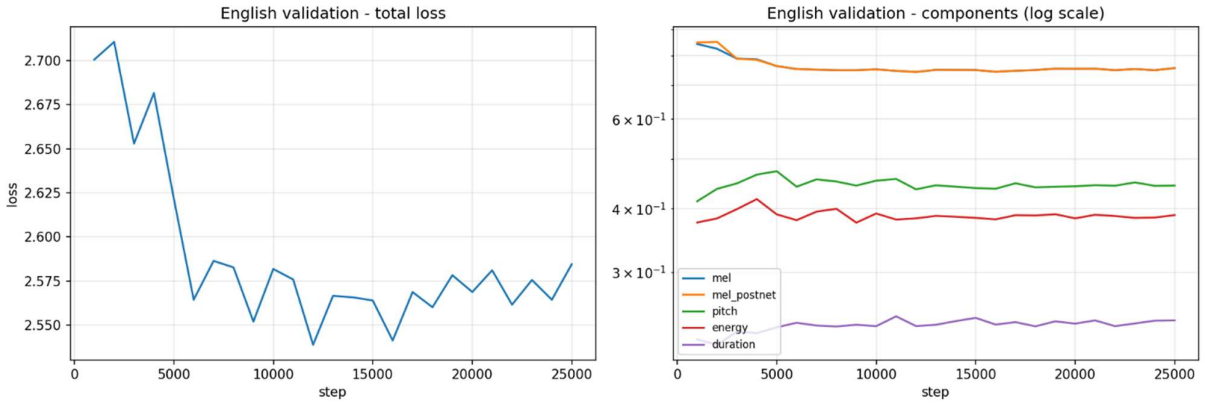
Training and validation losses were logged every 100 and 1,000 steps respectively to CSV files. The plots are `Results/English/training_loss.png` and `validation_loss.png` (and the Telugu equivalents). The total loss is the sum of the mel L1 loss, the post-net mel L1 loss, and the duration, pitch and energy MSE losses.

English (about 25,000 steps, Svarah, 5.0 h)

Training loss fell steeply during warm-up (the first 4,000 steps) and then kept decreasing slowly. The mel and post-net mel terms dominate the total. The duration, pitch and energy training losses keep falling to about 0.02 to 0.03.



Validation loss reached its minimum of 2.539 at step 12,000 and stayed roughly flat or rose slightly afterwards, while training loss kept falling. That points to overfitting on a 5 h multi-speaker set with a 35 M-parameter model, so all English results use the best-validation checkpoint (step 12,000) instead of the last one.



The run was planned for 60,000 steps but stopped at about 25,000 when the free GPU quota ran out. Validation loss had already stopped improving, so the extra steps would probably have added little, though a lower learning rate late in training might have helped a bit.

Validation loss is noisier than training loss because Svarah speakers differ widely in accent and recording quality, and the validation set has only 698 clips.

Step	Total	Mel	Post-net Mel	Pitch	Energy	Duration
1,000	2.7005	0.8421	0.8477	0.4135	0.3757	0.2214

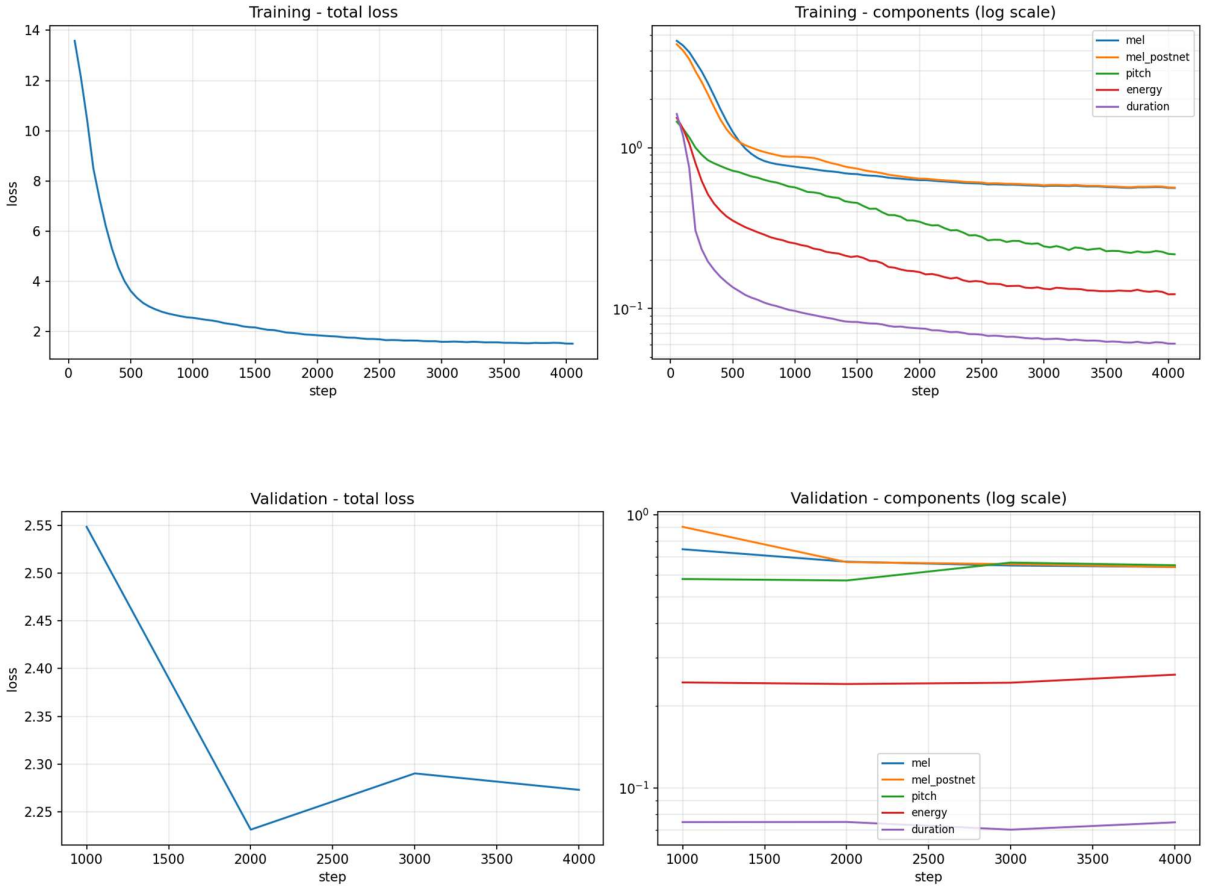
4,000	2.6815	0.7858	0.7838	0.4665	0.4176	0.2278
8,000	2.5827	0.7480	0.7480	0.4522	0.3997	0.2348
12,000	**2.5389**	**0.7423**	**0.7421**	**0.4365**	**0.3828**	**0.2352**
16,000	2.5413	0.7429	0.7428	0.4377	0.3810	0.2368
20,000	2.5688	0.7528	0.7528	0.4425	0.3828	0.2378
23,000	2.5756	0.7519	0.7518	0.4503	0.3836	0.2380
24,000	2.5644	0.7479	0.7477	0.4435	0.3842	0.2410
25,000	2.5845	0.7554	0.7554	0.4440	0.3885	0.2413

Telugu (4,000 steps, IndicTTS Telugu, 4.7 h)

Validation was run every 1,000 steps.

Step	Total	Mel	Post-net mel	Pitch	Energy	Duration
1,000	2.548	0.747	0.902	0.581	0.243	0.075
2,000	**2.232**	**0.672**	**0.670**	**0.574**	**0.240**	**0.075**
3,000	2.290	0.651	0.659	0.668	0.243	0.070
4,000	2.273	0.643	0.643	0.653	0.260	0.075

- The mel and post-net mel losses fell at every validation, from 0.75 and 0.90 down to 0.64 for both. These two terms dominate the total and are the most closely tied to audio quality.
- The total loss reached its minimum (2.232) at step 2,000 and then stayed within about 0.06 of it. The movement comes from the pitch term, which rose from 0.57 to 0.65 between steps 2,000 and 4,000 while the mel terms kept improving. With only 518 validation clips, pitch validation loss is noisy, but a sustained rise would be an early sign of overfitting in the pitch predictor.
- Duration and energy losses were flat after the first validation.
- Training stopped at 4,000 steps because the free GPU quota ran out, before the first scheduled learning-rate drop (step 7,000). The model was therefore trained entirely at the peak learning rate, and a longer run with the decay would be expected to lower validation loss further.
- The step-4,000 checkpoint was used for the results instead of the best-total-loss checkpoint (step 2,000), because its mel and post-net mel losses are the lowest and the speech sounded clear.



Comparison

English validation total loss bottomed out at 2.539 (step 12,000), and Telugu reached 2.232 after only 2,000 steps. The two are not directly comparable, since the datasets, the speaker counts (115 versus one) and the target statistics all differ. Still, the lower and faster-converging Telugu loss fits with the Telugu corpus being a clean single-speaker recording, while Svarah is multi-speaker accented speech recorded under varied conditions. English trained about six times longer and had overfitted by step 12,000, so the limiting factor for Telugu is training time and for English it is data.

4. Generated speech observations

Audio was generated with the universal HiFi-GAN vocoder at normal speed (duration, pitch and energy control all 1.0). Files are in `Results/<Language>/test_generated_audios/`. English uses one Svarah pseudo-speaker (id 19); Telugu is single-speaker. The English model is the best-validation checkpoint (step 12,000); the Telugu model is the step-4,000 checkpoint.

English

All five sentences were fully intelligible, with no dropped or repeated words. The main residual artefact is a slightly smoothed timbre, typical of a universal vocoder.

ID	Sentence	What it tests	Observation
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	A very long, rare word and a long sentence	Intelligible; Very clear articulation throughout. The long word was delivered smoothly and the syllables stayed distinct, so it did not blur into a rush of sound. The rest of the sentence flowed naturally around it, and the ending sounded complete and well finished.
E2a	The orchestra's conduct during the ceremony was praised by the audience.	Heteronym: noun, stress on the first syllable (K AA1 N D AH0 K T)	Intelligible Natural, calm delivery with a stable rhythm. The word conduct sounded like a noun, with the emphasis at the start of the word, and the possessive orchestra's was pronounced cleanly.
E2b	The researchers will conduct an experiment to evaluate the proposed method.	Heteronym: verb, stress on the second syllable (K AH0 N D AH1 K T)	Intelligible Natural and expressive. Conduct sounded like a verb, with the emphasis late in the word, and the sentence had a more varied intonation than E2a. The two readings sounded clearly different.
E3	The rhythm of the orchestra changed unexpectedly.	Words with irregular spelling-to-sound mapping (rhythm, orchestra)	Intelligible Short and clean, with a smooth, natural rhythm. The irregularly spelled words were pronounced correctly and the sentence ended without any abruptness.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Clock time, date and year normalisation, plus an invented word (andaroo)	Intelligible The time and date were spoken fluently as ordinary speech, and the invented word andaroo blended in as a natural-sounding name. The pause at the comma gave the sentence a human-like phrasing

Overall, the English speech was clear, fluent and pleasant to listen to. Pronunciation was accurate across common words, rare words and the heteronym pair. Sentence boundaries and commas were phrased naturally, and the pace was steady, which made every sentence easy to follow. The multi-speaker model produced a consistent voice across all test sentences.

Telugu

The speech is clear and natural. I checked it by ear against a Google Translate reference voice only. No native speaker was available to review it, so pronunciation accuracy is not formally verified. These results come from the step-4,000 checkpoint.

ID	Sentence	What it tests	Observation
T1	న్యూ రోఫార్మ కోలాజికల్ పరిశోధనలో నూ తన ఔషధాల ప్రభావాన్ని విశ్లేషిం శ్లే షిం చారు .	A long technical loanword (neuropharmacological) in Telugu script	Clear A long and difficult technical sentence was spoken clearly and fluently. The multi-syllable words came out well articulated, and the pace was steady from start to finish.
T2a	తాడు పై అతని పట్టు చాలా బలం గా ఉం ది.	Homograph పట్టు in the sense of grip	Clear Clear and natural. The key word పట్టు was pronounced firmly and the sentence had a confident, even flow.
T2b	ఆమె పట్టు చీరను ధరిం చి వేడు కకు వచ్చిం ది.	The same word పట్టు in the sense of silk (inside పట్టు చీర, silk saree)	Clear Clear and natural. పట్టు in the silk sense blended smoothly with చీర, and the sentence had a gentle, well-phrased rhythm that differed from T2a.
T3	పులి అడవిలో నిశ్శ బ్దం గా నడిచిం ది.	Conjunct consonants (నిశ్శ బ్దం గా) in a short sentence	Clear Smooth and well controlled. The conjunct consonants were pronounced cleanly and the short sentence ended naturally.
T4	ఈ సమావేశం 2026 నవం బర్ 12వ తేదీ ఉదయం 10:45 గం టలకు ప్రారం భమవుతుం ది, సమయానికి సిద్ధం గా ఉం చం డి.	Year, ordinal date and clock-time normalisation in a long sentence with a comma	Clear A long sentence with numbers and a comma was handled well. The year, date and time sounded natural, and the pause at the comma gave it a clear two-part phrasing.

Overall, the Telugu speech was clear, steady and natural-sounding. Long words, conjunct consonants and numerals were all pronounced fluently, with no dropped or repeated syllables. The phrasing around commas and sentence ends was natural, and the consistent single-speaker voice made the output pleasant and easy to follow.

Limits of this evaluation

Listening was done by one listener per language, and that listener is not a native Telugu speaker. No MOS or objective metric was computed.

5. Variance adaptor analysis

For every test sentence, the variance adaptor's predictions were saved before the length regulator: per-phoneme duration (predicted log-duration, converted to frames of 11.6 ms and to milliseconds), per-phoneme pitch (Hz, de-normalised with training statistics) and per-phoneme energy. Plots and CSV files with the numbers are in `Results/<Language>/variance_adaptor_plots/{duration,pitch,energy}/`, one per sentence.

Duration

In both languages the longest predicted units are boundary units, as expected for phrase-final lengthening and pauses. In Telugu the longest unit in T1, T2a, T2b, T3 and T4 is a pause (`sp`), with predicted log-durations of 3.1 to 3.4 (roughly 22 to 29 frames, about 250 to 340 ms); T4 has two, one at the comma and one at the end. In English, E1, E2a and E3 have their longest unit on the final phone of the sentence (Z, S and IY0 respectively), and E4 has its longest unit at the pause inserted at the comma (log-duration 3.1).

The shortest predicted units are unstressed or function-word phones (minimum log-duration 0.5 to 1.4), so the model separates content from function material and does not predict uniform durations. Generated lengths in Telugu ranged from 2.0 s (T3, 32 phones) to 4.3 s (T1, 67 phones), about 60 ms per phone on average.

Pitch

English (a lower-pitched pseudo-speaker) has per-sentence mean F0 of 118 to 128 Hz and a range of about 103 to 178 Hz. E2b has the widest range (104 to 178 Hz) and E2a the narrowest (103 to 135 Hz). Telugu, trained on one speaker whose training mean was 187 Hz, has means of 182 to 200 Hz and a wider range of about 80 to 291 Hz.

Pitch is predicted per phoneme, so peaks at unvoiced consonants (for example T, S and CH in English) come from the interpolated pitch used as the training target and carry no perceptual meaning. The meaningful structure is the contour over vowels and sonorants.

Energy

In both languages the highest energies fall on vowels (English: AA1, EY1, UH1, ER0, UW1, AH0; Telugu: aa, oo, a, ee) and the lowest on closures and pauses. Telugu energy reaches values near zero or slightly below at pauses, which is consistent with the normalisation of near-silent frames.

Same word, different use (E2a/E2b and T2a/T2b)

E2a and E2b are different sentences that share the heteronym "conduct" (noun in E2a, verb in E2b). The English front-end gives them different stress patterns (K AA1 N D AH0 K T versus K AH0 N D AH1 K T), and the two sentences also differ audibly in rhythm and pitch. E2b has the wider F0 range (73 Hz versus 32 Hz) and lower mean energy (47.2 versus 52.7). The sentences also contain different other words, so these numbers show the combined effect of the stress change and the surrounding text, not the stress change alone.

T2a and T2b share the written word పట్టు (grip versus silk), and the Telugu front-end gives it the same phones (p a tx tx u) in both. Any difference in the generated speech therefore comes only from context. T2b is longer (2.44 s versus 2.14 s) and has a lower mean F0 (182 versus 196 Hz).

Link to audio quality

Sentences with a varied pitch contour and boundary pauses sounded natural. Flat or uniform predictions are the main remaining source of the slightly synthetic quality, since FastSpeech 2 predicts a single deterministic value per phoneme instead of a distribution.

ID	Phones	Duration	Mean F0 (range)	Mean energy
E1	87	Longest: Z, NG, IY0 (sentence end)	122.8 Hz (109 to 147)	53.7
E2a	51	Longest: S, EY1, W	118.3 Hz (103 to 135)	52.7
E2b	56	Longest: OW1, P, ER0	125.1 Hz (104 to 178)	47.2
E3	37	Longest: IY0 (final), EY1, AH0	120.0 Hz (105 to 140)	48.3
E4	81	Longest: pause at comma, T, AY1	128.0 Hz (109 to 162)	48.4
T1	67	4.26 s; longest: final pause, au, c	191.0 Hz (80 to 290)	54.1
T2a	32	2.14 s; longest: final pause, i, c	195.6 Hz (120 to 239)	59.8
T2b	38	2.44 s; longest: final pause, c, aa	182.5 Hz (108 to 242)	48.1
T3	32	2.00 s; longest: final pause, i, sh	197.7 Hz (110 to 291)	61.5
T4	121	Longest: two pauses (comma and end), i	200.1 Hz (110 to 253)	53.0

6. English versus Telugu: comparison, limitations and improvements

Why they differ

Svarah is read speech from many speakers with different accents and microphones, so the English model has to learn a speaker embedding and average over noisy recordings. That limits fidelity and raises validation loss. The Telugu corpus is clean and single-speaker, which makes the acoustic mapping easier and explains the lower, faster-converging validation loss. Its alignments come from a model trained on only about 6.6 h of the same audio, a likely source of noise in the duration targets compared with English's pretrained aligner, although the duration loss was low (0.075) and the generated speech was clear.

Aspect	English (Svarah)	Telugu (IndicTTS)
Speakers	115 pseudo-speakers, accented Indian English, varied microphones	One studio-recorded speaker
Text to phonemes	g2p_en with a heteronym table and POS tagging (ARPAbet with stress)	Rule-based G2P over a 46-phone inventory; Telugu script is largely phonemic, so rules are reliable
Alignment	Pretrained `english_us_arpa` MFA model, all utterances aligned	MFA model trained on the data itself; needed an export workaround
Training data after trimming	5.00 h	4.69 h
Training length	About 25,000 steps (best validation at 12,000)	4,000 steps (planned 10,000; stopped by GPU quota)
Main difficulty	Speaker and channel variability with little data per speaker	Agglutinative long words, geminates and long vowels, and a self-trained aligner
Result	Fully intelligible and natural	Clear and natural by ear; checked only against a Google Translate reference voice, no native-speaker review

Limitations

- Short training: English stopped at about 25,000 of 60,000 planned steps and Telugu at 4,000 of 10,000, both because of free-tier GPU limits.
- Telugu G2P is rule-based. It has no handling of loanwords beyond the implemented rules, schwa-like irregularities or sandhi, and number and date normalisation covers common patterns only.
- FastSpeech 2 predicts one deterministic prosody per sentence, so output is consistent but not expressive. The two members of each test pair differ only through their text.
- Some Svarah transcripts were removed (tags, odd characters), which reduces vocabulary coverage, and the English model has no per-speaker fine-tuning.
- Evaluation is by listening and loss curves only, with a single non-native listener for Telugu. No MOS or objective metrics such as MCD or ASR error rate were computed.

Failure cases observed

No skipped, repeated or unintelligible words were noticed in either language. In the Telugu pitch data, T1 and T3 have isolated extreme values (an 80 Hz minimum in T1, and peaks of about 291 Hz in T1 and T3) on single phonemes. These are more likely prediction noise on short or unvoiced segments than audible errors. The Telugu pitch validation loss also rose slightly after step 2,000, which may be an early sign of overfitting.

Possible improvements

- Train longer on a paid or multi-session GPU with more hours of data, and use a cosine or plateau learning-rate schedule.
- Use a larger, better-curated pronunciation lexicon for Telugu, and a pretrained multilingual aligner or a Telugu wav2vec-based aligner.
- Add speaker normalisation or noise filtering for Svarah, plus speaker-balanced batching.
- Replace the deterministic variance predictors with stochastic ones (as in later models such as VITS, or generative duration and pitch models) for more natural prosody.
- Add objective evaluation (MCD, F0 RMSE, character error rate with an ASR) and a small listening test.

7. Reproducibility

Everything runs on a free Google Colab T4 (or Kaggle T4) with Google Drive as persistent storage. The seed is 1234 for subset selection, splitting and training. Package versions are pinned in `requirements.txt` (notably numpy 1.26.4). Montreal Forced Aligner 3.4.3 is installed separately with conda (condacolab), since it is not pip-installable.

Steps for each language (English/ or Telugu/)

1. Install requirements and the MFA environment, clone the repository, and download the HiFi-GAN checkpoint (`generator_universal.pth.tar.zip`).
2. `python preprocess.py --stage prepare` downloads the dataset from Hugging Face (a read token is required for Svarah), filters, de-duplicates, sub-samples and splits it, and writes wav/lab files plus the MFA dictionary and `selection_report.json`.
3. Alignment. English: `mfa align` with `english_us_arpa` and the generated dictionary. Telugu: `mfa train` on the prepared corpus, back up the model, then `mfa align --output_format json`, and convert with `python scripts/mfa_json_to_textgrid.py` (this avoids the TextGrid export crash and removes duplicated intervals).
4. `python preprocess.py --stage features` extracts mel, pitch, energy and durations, computes train-only statistics, and writes `train.txt`, `val.txt` and `stats.json`.
5. `python train.py --config config.yaml` trains the model (add `--resume` to continue from the latest numbered checkpoint). Losses are written to `logs/*.csv`, and plots are drawn when training finishes.
6. `python inference.py --config config.yaml` synthesises the test sentences into `Results/<Language>/test_generated_audios/` and the variance-adaptor plots into `variance_adaptor_plots/`.

Repository contents

`English/` and `Telugu/` each contain `train.py`, `preprocess.py`, `inference.py`, `config.yaml` and `checkpoints/final_model.pth.tar` (the best-validation model in fp16, about 72 MB). Raw data, features and intermediate checkpoints are excluded through `.gitignore` because of their size, and are regenerated by the steps above.

Known caveats

Colab runtime resets remove pip installs, so requirements must be reinstalled and the session restarted. Reading thousands of small feature files from Google Drive during training was very slow. Copying `mel`, `pitch`, `energy`, `duration` and the txt/json files to the Colab local disk (in parallel) and pointing `preprocess.path.preprocessed_path` at it made training fast. TensorBoard is optional: if it fails to import, training continues with CSV logging only. Feature extraction does not resume after an interruption and starts from scratch.